

9. What is the approximate number of atoms in a cubic metre of an ideal monatomic gas at the temperature of 27°C and a pressure of $1 \times 10^5 \text{ Pa}$?

A 1×10^{22}

B 6×10^{23}

C 2×10^{25}

D 3×10^{26}

Ans: C

$$pV = NkT$$

$$(1 \times 10^5)(1) = N(1.38 \times 10^{-23})(27 + 273.15)$$

$$N = 2.41 \times 10^{25}$$

$$\approx 2 \times 10^{25}$$

10. A fixed mass of monatomic ideal gas at a volume V at temperature T and pressure P